

Machine Learning with SVM

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Introduction

This report explores the use of a Support Vector Machine (SVM) with a Polynomial Kernel to classify the Iris dataset. It demonstrates data splitting, model training, and evaluation.

Data Import

We start by importing the Iris dataset that is in-built with in R (The dataset is often used for classification tasks to predict the species of the flower based on the four continuous measurements (sepal and petal length/width))

```
# Importing the iris dataset  
data("iris")  
  
# View the dataset  
head(iris)
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species  
## 1         5.1         3.5         1.4         0.2  setosa  
## 2         4.9         3.0         1.4         0.2  setosa  
## 3         4.7         3.2         1.3         0.2  setosa  
## 4         4.6         3.1         1.5         0.2  setosa  
## 5         5.0         3.6         1.4         0.2  setosa  
## 6         5.4         3.9         1.7         0.4  setosa
```

Checking for Missing Data

Before proceeding with the analysis, we check if there are any missing values in the dataset.

```
## [1] 0
```

Data Splitting

We set a random seed for reproducibility and split the data into training and testing datasets. 80% of the data is used for training, and the remaining 20% is used for testing.

```
# Setting a random seed number for reproducibility

set.seed(100)

# Performing stratified random split of the dataset

TrainingIndex <- createDataPartition(iris$Species, p=0.8, list = FALSE)
TrainingSet <- iris[TrainingIndex, ] # Training set
TestSet <- iris[-TrainingIndex, ] # Test set
```

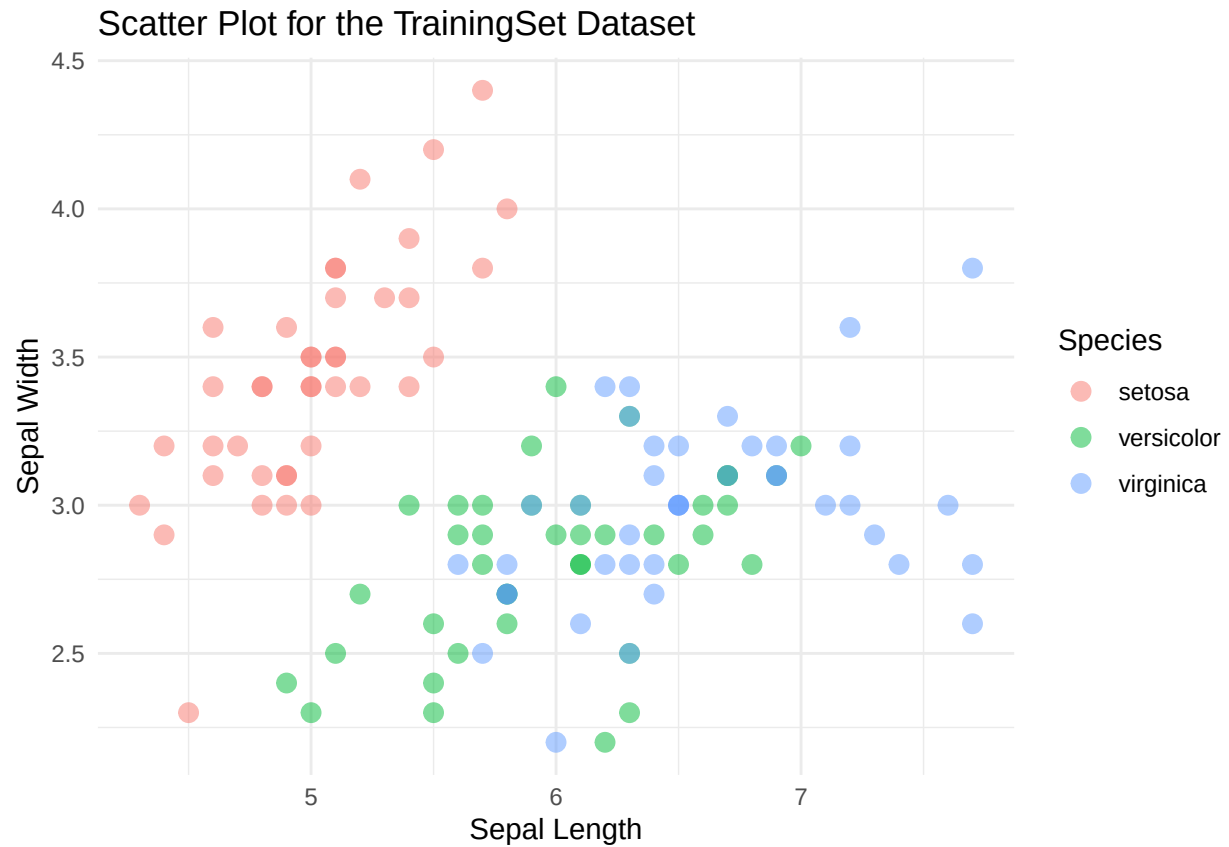
Visualizations

We use scatter plots and line graphs to visualize the data in both the training and test sets.

The scatter plot highlights that sepal dimensions can be a strong differentiator for setosa, as its cluster is well-separated. However, there is moderate overlap between versicolor and virginica, suggesting that additional features such as petal dimensions may be needed for accurate classification between these two species.

```
# Using the scatter plot to visualize the TrainingSet dataset

TrainingSet %>%
  ggplot(aes(x = Sepal.Length, y = Sepal.Width, colour = Species)) +
  geom_point(size = 3, alpha = 0.5) +
  labs(title = "Scatter Plot for the TrainingSet Dataset", x = "Sepal Length",
        y = "Sepal Width", color = "Species") +
  theme_minimal()
```

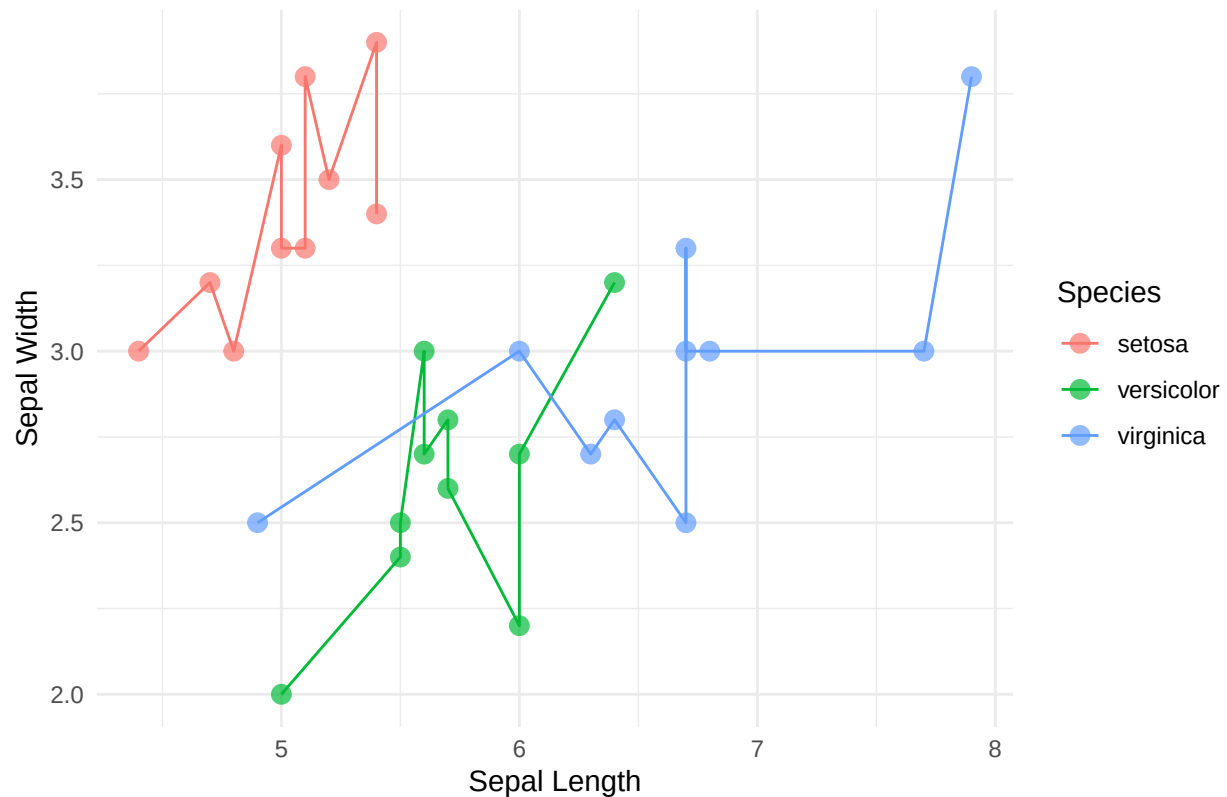


Test Set Visualization

```
# Using the line graph to visualize the TestSet dataset

TestSet %>%
  ggplot(aes(x = Sepal.Length, y = Sepal.Width, colour = Species)) +
  geom_point(size = 3, alpha = 0.7) +
  labs(title = "Scatter Plot for the TestSet Dataset", x = "Sepal Length",
       y = "Sepal Width", color = "Species") +
  geom_line() +
  theme_minimal()
```

Scatter Plot for the TestSet Dataset



Model Training

We train an SVM model with a polynomial kernel using the training dataset.

```
# Defining the training model using SVM model - Polynomial Kernel
Model <- train(
  Species ~ .,
  data = TrainingSet,
  method = "svmPoly",
  na.action = na.omit,
  preProcess = c("scale", "center"),
  trControl = trainControl(method = "none"),
  tuneGrid = data.frame(degree = 1, scale = 1, C = 1)
)
```

Cross-validation Model

We build a cross-validation model to evaluate the performance more rigorously. The model performed very well, achieving high accuracy of 97.5% and Kappa value of 96.3%, indicating strong predictive performance and agreement between predicted and actual classes. The tuning parameters were fixed, meaning no hyper-parameter tuning was performed during cross-validation.

```
# Building the CV model
Model.cv <- train(
  Species ~ .,
  data = TrainingSet,
  method = "svmPoly",
  na.action = na.omit,
  preProcess = c("scale", "center"),
  trControl = trainControl(method = "CV", number = 10),
  tuneGrid = data.frame(degree = 1, scale = 1, C = 1)
)
```

Predictions and Model Evaluation

We make predictions on both the training and test datasets and evaluate the model's performance.

```
# Apply the model for prediction
Model.training <- predict(Model, TrainingSet) # Applying the model to make predictions on the training
Model.testing <- predict(Model, TestSet)      # Applying the model to make predictions on the test data
Model.cv <- predict(Model.cv, TrainingSet)    # Perform cross-validation
```

```
# Model Performance (Display Confusion matrix and Statistics)
Model.training.confusion <- confusionMatrix(Model.training, TrainingSet$Species)
Model.testing.confusion <- confusionMatrix(Model.testing, TestSet$Species)
Model.cv.confusion <- confusionMatrix(Model.cv, TrainingSet$Species)
```

Confusion matrix for the training model

- All setosa 40 samples were correctly classified.
- 40 Versicolor samples were correctly classified, with 1 misclassified as virginica.
- 39 Virginica samples were correctly classified.

The model demonstrates excellent performance with high accuracy of 99.2% and Kappa value of 98.8%. The confusion matrix shows minimal misclassification, and the class-specific statistics indicate strong sensitivity, specificity, and predictive values across all classes.

```
# Printing the model performance
print(Model.training.confusion)
```

```
## Confusion Matrix and Statistics
##
##              Reference
## Prediction  setosa versicolor virginica
##   setosa      40           0           0
##   versicolor   0           40           1
##   virginica    0           0          39
##
## Overall Statistics
##
##              Accuracy : 0.9917
```

```
##                      95% CI : (0.9544, 0.9998)
##      No Information Rate : 0.3333
##      P-Value [Acc > NIR] : < 2.2e-16
##
##                      Kappa : 0.9875
##
##      McNemar's Test P-Value : NA
##
## Statistics by Class:
##
##                      Class: setosa Class: versicolor Class: virginica
## Sensitivity                1.0000                1.0000                0.9750
## Specificity                1.0000                0.9875                1.0000
## Pos Pred Value             1.0000                0.9756                1.0000
## Neg Pred Value             1.0000                1.0000                0.9877
## Prevalence                 0.3333                0.3333                0.3333
## Detection Rate             0.3333                0.3333                0.3250
## Detection Prevalence       0.3333                0.3417                0.3250
## Balanced Accuracy          1.0000                0.9938                0.9875
```

Confusion matrix for the testing model

- 10 Setosa samples were correctly classified.
- 9 Versicolor samples were correctly classified, with 1 misclassified as virginica.
- 10 Virginica samples were correctly classified

The model performed very well on the testing set, achieving high accuracy of 96.67% and Kappa value of 95%. The confusion matrix shows minimal misclassification, with only one versicolor sample misclassified as virginica. The overall statistics indicate strong predictive performance and agreement between predicted and actual classes.

```
print(Model.testing.confusion)
```

```
## Confusion Matrix and Statistics
##
##                      Reference
## Prediction  setosa versicolor virginica
##   setosa      10           0           0
##   versicolor   0           9           0
##   virginica    0           1          10
##
## Overall Statistics
##
##                      Accuracy : 0.9667
##                      95% CI : (0.8278, 0.9992)
##      No Information Rate : 0.3333
##      P-Value [Acc > NIR] : 2.963e-13
##
##                      Kappa : 0.95
##
##      McNemar's Test P-Value : NA
##
```

```
## Statistics by Class:
##
##           Class: setosa Class: versicolor Class: virginica
## Sensitivity           1.0000           0.9000           1.0000
## Specificity           1.0000           1.0000           0.9500
## Pos Pred Value        1.0000           1.0000           0.9091
## Neg Pred Value        1.0000           0.9524           1.0000
## Prevalence            0.3333           0.3333           0.3333
## Detection Rate        0.3333           0.3000           0.3333
## Detection Prevalence  0.3333           0.3000           0.3667
## Balanced Accuracy      1.0000           0.9500           0.9750
```

Confusion matrix for the cross validation model

- Setosa wa Correctly predicted 40 times.
- Versicolor was Correctly predicted 40 times, with 1 misclassified as Virginica.
- Virginica was Correctly predicted 39 times

This indicates that your model is performing exceptionally well with high accuracy and a strong Kappa value, suggesting excellent agreement between the predicted and actual classifications.

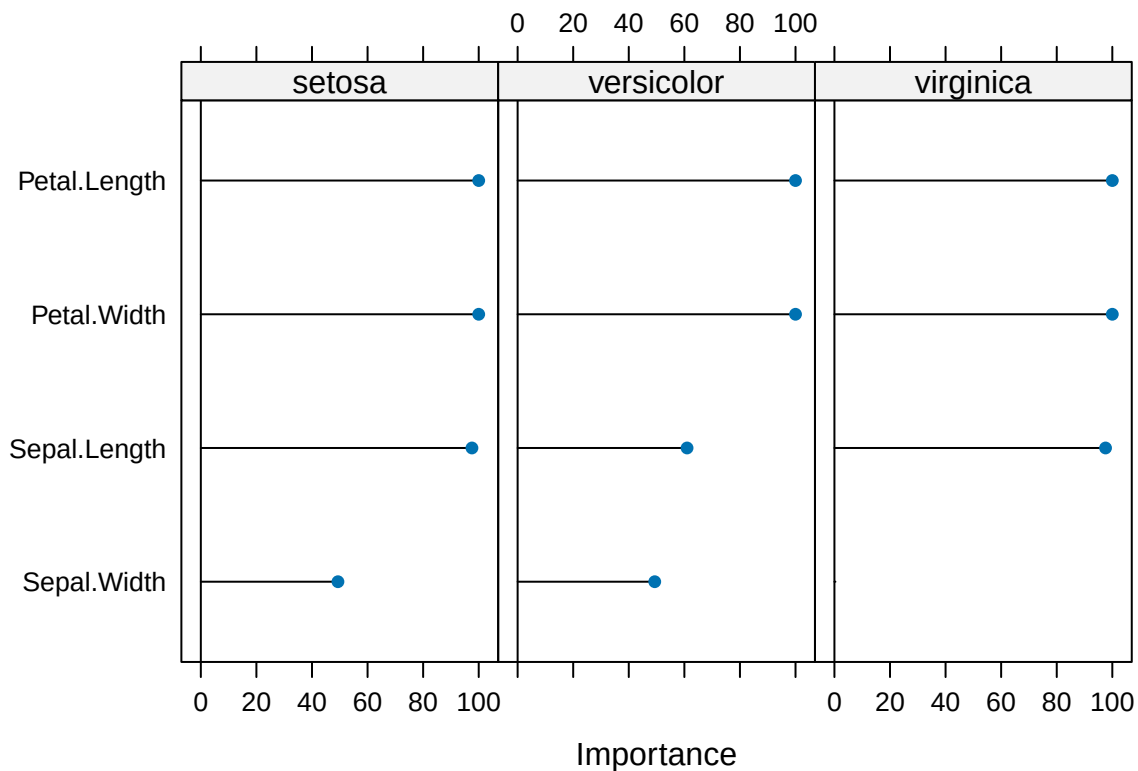
```
print(Model.cv.confusion)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  setosa versicolor virginica
##   setosa      40           0           0
##   versicolor   0          40           1
##   virginica    0           0          39
##
## Overall Statistics
##
##           Accuracy : 0.9917
##           95% CI : (0.9544, 0.9998)
##   No Information Rate : 0.3333
##   P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.9875
##
##   McNemar's Test P-Value : NA
##
## Statistics by Class:
##
##           Class: setosa Class: versicolor Class: virginica
## Sensitivity           1.0000           1.0000           0.9750
## Specificity           1.0000           0.9875           1.0000
## Pos Pred Value        1.0000           0.9756           1.0000
## Neg Pred Value        1.0000           1.0000           0.9877
## Prevalence            0.3333           0.3333           0.3333
## Detection Rate        0.3333           0.3333           0.3250
## Detection Prevalence  0.3333           0.3417           0.3250
## Balanced Accuracy      1.0000           0.9938           0.9875
```

Feature Importance

We evaluate the feature importance to understand which variables contribute the most to the model. - Petal.Length and Petal.Width have the highest importance score of 100 across all three classes. This indicates they are the most crucial variables for distinguishing between the different species of iris flowers. - Sepal.Length has high importance, especially for setosa and virginica, with scores of 97.56. However, its importance drops to 60.98 for versicolor, suggesting it is less critical for distinguishing this class compared to the others. - Sepal.Width is moderately important for setosa and versicolor with scores of 49.34 but has no importance (0.00) for virginica. This suggests that Sepal.Width does not contribute to distinguishing virginica from the other species.

```
# Feature Importance
Importance <- varImp(Model)
plot(Importance)
```



```
plot(Importance, col = "Red")
```